

1(a)(i)	<i>independent variable</i> temperature ; <i>dependent variable</i> volume of carbon dioxide / time to collect stated or fixed volume of CO ₂ ;	2
1(a)(ii)	<i>idea that:</i> ensures glucose and yeast are, uniformly distributed / homogeneously mixed / evenly mixed / completely mixed / AW ;	1
1(a)(iii)	<i>any seven from:</i> 1 use a minimum of five different temperatures ; 2 state a suitable range of identified temperatures ; 3 <i>idea of</i> same / constant / stated / standardised / known, volume of, yeast-glucose / mixture (in the test-tube) ; 4 method of maintaining a, constant / stated, temperature ; 5 leave mixture to equilibrate to (each) temperature ; 6 leave for, set time / stated time, and measure volume (of carbon dioxide) or time how long to collect same volume (of carbon dioxide) from (each) sample ; 7 use a control using, killed yeast / water replacing yeast ; 8 <i>ref. to</i> a minimum of, two repeats / three replicates (for each temperature), and calculating means ; 9 <i>idea of</i> repeat investigation with narrower range of temperatures (around that which seems the one giving maximum rate) ; 10 low risk ; 'medium risk' needs hazard and risk and precaution. A yeast + allergy/irritant + gloves / mask / goggles /PPE 11 AVP e.g. maintaining a constant pH and using a buffer ;	7

Question	Answer	Marks
1(a)(iv)	1 horizontal axis labelled temperature / °C and vertical axis labelled volume of carbon dioxide / cm³ / mm³ ; 2 curve starts low, reaches a maximum and then decreases ;	2
1(b)	1 flask 2 (provides the evidence) as in <i>C. tropicalis</i> there is lower CO₂ (production / volume / AW) ; 2 (for flask 2, <i>C. tropicalis</i> compared to <i>S. cerevisiae</i>) 65% v 93% (after 30 minutes) CO₂ as % of maximum or manipulated figure i.e. 28% different ; 3 <i>idea that</i>: when transferred from conditions with oxygen to conditions with no oxygen yeasts have to use fermentation / anaerobic respiration but <i>C. tropicalis</i> does so at a much slower rate (as no Crabtree effect) thus less CO₂ produced ora ;	3